PAPER – 4: COST AND MANAGEMENT ACCOUNTING

QUESTIONS

Case Scenario 1

EcoTrans Logistics Pvt. Ltd. is a regional goods transport company based in the industrial hub of LogiPort, providing daily freight services to and from GreenCity, a fast-growing metropolitan area located 120 kilometers away. The company has been operating for over a decade and has built a reputation for timely deliveries, especially for fast-moving consumer goods (FMCGs), packaged food items, and light industrial components.

To meet increasing demand, EcoTrans has recently expanded its fleet and now operates 24 transport vehicles of varying capacities. The company promises reliable daily service, and each vehicle typically completes one round trip per day, carrying cargo in both directions. EcoTrans emphasizes sustainability and efficiency, with systems in place for fuel monitoring, vehicle maintenance, and cost control.

However, to remain competitive and ensure profitability in a cost-sensitive industry, management has decided to undertake a detailed cost analysis for the month of April 2025. This analysis will guide pricing strategy and assess whether current operations are yielding the desired profit margin of 20% on freight revenue.

The following data has been compiled by the Finance and Operations departments for cost analysis and pricing decision-making.

Fleet Composition:

No. of Vehicles	Capacity per Vehicle (MT)
6	10 MT
8	14 MT
6	18 MT
4	22 MT

Each vehicle operates one round trip per day on average. Vehicles are loaded to 85% capacity on the forward journey and 65% on the return. On average, 12% of the fleet is under maintenance at any time.

The company operated for 26 days in April 2025.

Monthly Expenditure (April 2025)

Description	Amount (₹)
Salary of Transport Manager	75,000
Salary of 32 Drivers @ ₹22,000 each	7,04,000
Wages of 28 Helpers @ ₹13,000 each	3,64,000
Loading & Unloading Charges per Trip	950
Consumable Stores (Fuel Additives, Tools)	1,50,000
Toll Charges	250 per Trip
Insurance (Annual Premium)	9,60,000
Road Licence Fee (Annual)	5,40,000
Diesel Cost per Litre	82
Mileage per Vehicle	4.5 km/litre
Lubricants & Oils	1,30,000
Tyres, Tubes, Spare Parts (Running basis)	4,80,000
GPS & Fleet Tracking Subscription (Monthly Basis)	2,000 per vehicle
Driver Safety & Training Programmes	80,000
Garage Rent (Annual)	10,80,000
Workshop Repairs (Routine & Emergency)	3,40,000

Electricity & Utility Expenses	62,000
Depreciation on Vehicles (Monthly)	6,80,000

The company operates an in-house workshop that handles routine and emergency repairs for both its own fleet and external vehicles. As part of the internal cost allocation, 45% of the Transport Manager's salary is charged to the workshop to reflect time and resources dedicated to overseeing maintenance operations. In return, the transport department has been apportioned ₹95,000 by the workshop for repair and service support during the month.

Being a finance manager of the company, you are required to answer the followings (5 MCQs):

(i) What is the total fixed cost for April 2025?

(A) ₹21,95,000

(B) ₹20,89,750

(C) ₹22,89,250

(D) ₹19,98,250

(ii) Calculate the total variable cost for April 2025

(A) ₹44,71,885

(B) ₹31,50,000

(C) ₹29,65,920

(D) ₹48,45,000

(iii) What should be the freight rate per ton-km to achieve a 20% profit margin?

(A) ₹4.461

(B) ₹5.576

(C) ₹4.372

(D) ₹3.845

(iv) Calculate the total wages paid to loading and unloading labour in April 2025.

(A) ₹10,43,520

(B) ₹10,42,560

(C) ₹9,85,000

(D) ₹10,37,400

(v) What is the total ton-kilometers (ton-km) generated by EcoTrans Logistics in April 2025?

(A) 1,31,789 ton-km

(B) 5,49,120 ton-km

(C) 15,15,571 ton-km

(D) 26,00,000 ton-km

Case Scenario 2

In order to provide employment opportunities in rural areas of Madhya Pradesh, State government decided to give interest free loan up to ₹ 30,00,00,000 to Khadi cloth industrialists. In return these industrialists are expected to set up the factory and produce khadi cloth in such rural areas.

The loan is refundable after 20 years in lump sum. Madhya Pradesh government kept a stipulation that 40% of cloth production shall be supplied to government at a nominal rate of ₹ 10 per metre. This supply shall be used by the government for making uniforms of class IV employees of state government.

Haryana cloth mill of Hissar took loan of ₹ 30 crores and commenced business in Dindori, a rural district of Madhya Pradesh. Production commenced at Dindori and following are the details:

Production during the year 2023-24	**40,00,000 metres of Khadi cloth**
Raw cotton used	₹ 1,20,00,000
Direct wages	₹ 1,80,00,000
Freight inwards	₹ 8,00,000
Factory overheads estimated at 30% of prime cost	
Office and administration overheads estimated at 20% of factory cost	

Donations	₹ 2,00,000
Fines & Penalties	₹ 5,00,000
Selling and distribution overhead estimated at	₹ 1,69,52,000

The selling price to be charged from general public is to be fixed in such a way as to earn a profit of 30% on total cost of sales and price should be in multiple of Re.1.

Actual overheads incurred are as under:-

(1) Factory overheads = ₹ 89,50,000

(2) Office and administration overheads = ₹ 90,50,000

(3) Selling and distribution overheads = ₹ 1,70,00,000

The company has been requested to submit a cost audit report to the government detailing the production activities in its factory. As part of this process, the government has provided a draft format containing several key questions that must be addressed within the report. These questions are designed to ensure the accuracy and transparency of the company's production costs, and they play a critical role in shaping the audit report. Before proceeding with the preparation of the final cost audit report, it is essential that the company carefully reviews and works out the answers to these questions.

(i) Determine cost of sales per metre of cloth produced?

(A) ₹ 13.25

(B) ₹ 16.25

(C) ₹ 18.25

(D) ₹ 15.25

(ii) Determine the loss element per metre in government supply?

(A) ₹ 7.25

(B) ₹ 5.75

(C) ₹ 4.50

(D) ₹ 6.25

(iii) What will be the selling price per metre charged from general public?

(A) ₹ 29

(B) ₹ 27

(C) ₹ 31

(D) ₹ 33

(iv) What will be the rate of return on overall sales proceeds as per financial accounts?

(A) 22.313%

(B) 18.781%

(C) 24.387%

(D) 20.564%

(v) What will be the net profit in financial accounts from the factory, if government fixed the price of the cloth to be sold to general public at ₹ 28?

(A) ₹ 1,67,20,000

(B) ₹ 1,66,00,000

(C) ₹ 1,66,20,000

(D) ₹ 1,67,00,000

3. A manufacturing company has set the standard cost for producing one unit of its product as follows:

- Direct Material: 10 kg @ ₹ 5/kg
- Direct Labour: 4 hours @ ₹ 20/hour
- Variable Overhead: 4 hours @ ₹ 5/hour
- Fixed Overhead: ₹ 10 per unit (based on normal output of 5,000 units)

During a particular month, the company produced 4,800 units. The actual results were:

- Direct Material Used: 49,000 kg costing ₹ 2,45,000

- Direct Labour: 19,200 hours costing ₹ 3,84,000
- Variable Overhead Incurred: ₹ 95,000
- Fixed Overhead Incurred: ₹ 52,000

Based on this information, which of the following statements is most accurate?

(A) The material usage variance is ₹ 5,000 (Favourable)

(B) The labour rate variance is ₹ 19,200 (Adverse)

(C) The labour rate variance is ₹ 19,200 (Favourable)

(D) The fixed overhead volume variance is ₹ 2,000 (Adverse)

4. A manufacturing firm has produced 10000 units of a standard product by incurring the following expenses:

Material costs	:	₹ 80,000
Labour costs	:	₹ 40,000
Production overhead	:	₹ 30,000
Office overhead	:	20% of factory cost

Estimated selling and distribution overhead is ₹ 4 per unit. If the firm has no inventory at the beginning of the period and expects to maintain 20% of output as closing inventory, at what price per unit the product should be sold to earn a profit of 20% on sales?

(A) ₹ 27.50

(B) ₹ 23.00

(C) ₹ 26.40

(D) ₹ 28.75

5. Product AB has a standard cost of ₹ 32, ₹ 6 of which relates to direct materials. Budgeted production for the month was 1600 units. During the month, 1500 units of AB were produced and materials worth ₹ 9,000 were purchased. There was no opening stock of materials but closing stock, which was valued at standard cost, amounted to ₹ 1,500. What is the total variance for materials?

(A) ₹ 500 (F)

(B) ₹ 500 (A)

(C) ₹ 1500 (F)

(D) ₹ 1500 (A)

6. Rinki Ltd. has two divisions, A and B. Target profit of A is ₹ 100 lakh. A has capacity to produce 20,000 units of product P but only 15,000 units are produced and sold in the market at ₹ 3,000 per unit. B received an order for which P would be required as input. B approaches A for purchase of 5000 units of P. What price should A charge from B per unit of P so as to meet its target profit? (Other details of A are: Fixed cost is ₹ 90 lakh; Variable cost per unit of P is ₹ 2,000)

(A) ₹ 3,000

(B) ₹ 2,800

(C) ₹ 2,500

(D) ₹ 2,000

7. A company plans to produce 44,000 units during the budget period. The company requires 4 units of raw material to produce 1 unit of finished goods. The company currently has 1,00,000 units of raw material on hand and wishes to have an inventory of 1,10,000 units of raw material on hand at the end of the budget period. The number of raw material units to be purchased during the budget period is

(A) 1,76,000 units

(B) 1,67,000 units

(C) 1,86,000 units

(D) 1,68,000 units

Material Cost

8. XYZ Enterprises manufactures different types of beverages. The monthly demand pattern for these beverages is as follows:

- Orange Juice: 60,000 litres
- Apple Juice: 20,000 litres
- Grape Juice: 5,000 litres

To produce one litre of Apple Juice, Orange Juice, and Grape Juice, 6 kg of apples, 3 kg of oranges, and 5 kg of grapes are required, respectively. There is no opening or closing stock of raw materials.

XYZ Enterprises can source the raw materials either from local farmers (import in case or grapes) or from wholesale suppliers. The company has the flexibility to purchase any quantity of materials from the wholesale suppliers, but when buying from the farmers or in case of import, it must adhere to a minimum quantity that is specified for each type of material. Below is a summary of the purchasing conditions:

Material	Wholesale Supplier	Farmers
Oranges	Minimum purchase: Any quantity	7,20,000 kg
	Price per kg: ₹ 18.00	Price per kg: ₹ 15.00
	Transportation: ₹ 5,000	Transportation: ₹ 20,000
	-	Sorting and piling cost: ₹ 1,500
	Loading cost: ₹ 10/50 kg	Loading cost: ₹5/50 kg
	Unloading cost: ₹ 2/50 kg	Unloading cost: ₹2/50 kg
Apples	Minimum purchase: Any quantity	3,60,000 kg
	Price per kg: ₹ 12.00	Price per kg: ₹ 10.00
	Transportation: ₹ 11,000	Transportation: ₹ 15,000
	-	Sorting and piling cost: ₹ 1,200
	Loading cost: ₹10/50 kg	Loading cost: ₹ 3/50 kg
	Unloading cost: ₹2/50 kg	Unloading cost: ₹ 2/50 kg
	Wholesale Supplier	**Import**
Grapes	Minimum purchase: Any quantity	1,50,000 kg
	Price per kg: ₹ 32.00	Price per kg: ₹ 25.00
	Transportation: ₹ 3,000	Transportation: ₹ 15,000
	Loading cost: ₹ 10/50 kg	Loading cost: ₹ 25/50 kg
	Unloading cost: ₹2/50 kg	Unloading cost: ₹2/50 kg

Transportation and Sorting and piling cost is given for per purchase order.

8% import duty is applicable is purchase of grapes for which ITC is not available.

XYZ Enterprises is also paying an annual interest rate of 15% on cash credit facility and ₹1,000 per 1,000 kg of rent to its warehouse.

You are required to:

1. Calculate the total purchase cost for each material (Apple, Orange, and Grape) under the following two options:
   (a) Purchase from Wholesale Supplier
   (b) Purchase from Farmers/Import
2. Calculate the Economic Order Quantity (EOQ) for each material under both options.
3. Provide a Recommendation on the best purchase option for the "Grapes" material if minimum purchase quantity for import is 1,50,000 kg.

Employee Cost

9. Tommy HL Ltd. pays the following to a skilled worker engaged in production works. The following are the employee benefits paid to the employee:

(a)	Basic salary per day	₹1,000
(b)	Dearness allowance (DA)	20% of basic salary
(c)	House rent allowance	16% of basic salary
(d)	Transport allowance	₹50 per day of actual work
(e)	Overtime	Twice the hourly rate (considers basic and DA), only if works more than 9 hours a day otherwise no overtime allowance. If works for more than 9 hours a day then overtime is considered after 8^{th} hours.

(f)	Work of holiday and Sunday	Double of per day basic rate provided works atleast 4 hours. The holiday and Sunday basic is eligible for all allowances and statutory deductions.
(h)	Earned leave & Casual leave	These are paid leave.
(h)	Employer's contribution to Provident fund	12% of basic and DA
(i)	Employer's contribution to Pension fund	7% of basic and DA

The company normally works 8-hour a day and 26-day in a month. The company provides 30 minutes lunch break in between.

During the month of August 2025, Mr. Shyam works for 23 days including 15th August and a Sunday and applied for 3 days of casual leave. On 15th August and Sunday he worked for 5 and 6 hours respectively without lunch break.

On 5th and 13th August he worked for 10 and 9 hours respectively.

During the month Mr. Shyam worked for 100 hours on Job no. AB100.

You are required to CALCULATE:

(i) Earnings per day

(ii) Effective wages rate per hour of Mr. Shyam.

(iii) Wages to be charged to Job no. AB100.

Overheads: Absorption Costing method

10. BrightTech Appliances Ltd. is a leading manufacturer of smart home devices such as the SmartCool AC, PowerLite Inverter, and EcoDry Dryer. Amid increasing operational complexity, the company's finance controller has initiated a review of overhead allocations to gain better control over departmental efficiency and product profitability.

The company operates through four major departments:

1. **Production** – handles manufacturing across three plants using advanced machinery including stores management.

2. **Administration** – supports HR, finance, legal, and compliance.
3. **Sales & Distribution** – manages retail tie-ups, e-commerce, and logistics.
4. **General Management** – oversees strategy, innovation, and executive-level operations.

The extract from the budget for the next financial year is as follows:

Particulars	Amount (₹)	Notes
Raw Material Cost	5,75,00,000	Consumed in a ratio of 3:4:3 for the three products
Indirect Material Cost	12,50,000	Manufacturing ₹7,20,000, Stores ₹25,000, General admin ₹3,80,000, Personnel ₹95,000, Sales ₹30,000
Salary & Wages	4,20,00,000	Includes direct wages of ₹1,60,00,000 (temporary contract workers)
Rent & Property Tax	₹1,20,000	₹90,000 Warehouse Rent, ₹30,000 Property Tax
Depreciation on Non-Current Assets	₹25,00,000	Factory/office building ₹10,50,000, ACs ₹2,00,000, Machinery ₹12,50,000
Power & Fuel	₹4,90,000	₹4,70,000 for manufacturing, ₹20,000 for delivery vans
Machinery Insurance Premium	₹5,00,000	At 4% of WDV of machinery
Group Employee Insurance	₹3,10,000	Based on gross salary (excluding direct workers and top management)
Printing & Stationery	₹8,20,000	Manufacturing ₹21,000, Finance ₹5,50,000, Legal ₹28,000, Sales ₹2,21,000
Audit Fees	₹1,40,000	Statutory and internal audit

Electricity Expense	₹4,00,000	Metered use: Production 5,600, Admin 10,000, Sales & Distribution 4,000, General Management 1,600 units
Telephone & Mobile	₹4,95,000	Production ₹1,25,000, Personnel ₹50,000, General Mgmt ₹30,000, Sales ₹75,000, Customer Support ₹2,15,000
Travelling Expenses	₹24,00,000	Production ₹5,50,000, General Mgmt ₹12,00,000, Sales ₹6,50,000
Meal Coupon Subsidy	₹2,25,000	₹3,000 per employee on roll
Software License Renewals	₹16,50,000	Stores Management ₹8,50,000, Finance ₹1,50,000, Stores ₹30,000, Customer Support ₹6,20,000
Misc. Expenditures	₹9,50,000	Allocated based on direct expenses

Additional Information:

- **Employee Count:** Production 22, Admin 20, Sales & Distribution 28
- **Average Gross Salary:** Production ₹6,30,000, Admin ₹4,20,000, Sales & Distribution ₹3,50,000, General Mgmt ₹2,80,000
- **Floor Area:** Production 9,900 m^2, Admin 3,600 m^2, Sales & Distribution 2,700 m^2, General Mgmt 1,800 m^2
- **AC Tonnage:** Production 6,000 RT, Admin 3,000 RT, S&D 3,000 RT, General Mgmt 1,500 RT
- **Depreciation Rates:** Building 5%, AC 15%, Machinery 10%
- **Department Roles:**
    - General Mgmt: 70% Sales strategy, 20% Production/ Marketing, 10% Admin
    - Admin: 50% Production, 50% Sales.

Required:

(i) Prepare a schedule of Cost Allocation for Production Dept., Administration Dept., Sales & Distribution Dept. and General Management.

(ii) Prepare a schedule of Cost Apportionment (Primary Distribution) for Production Dept., Administration Dept., Sales & Distribution Dept. and General Management.

(iii) Prepare a schedule of Secondary distribution of Administration Dept. and General Management costs.

Activity Based Costing

11. Edward Ltd. manufactures weighing machines of standard size and sells its products to two industrial customers namely MT Ltd. and KG Ltd. and to a dealer MG Bros. having shops in different cities. The maximum retail price per unit of weighing machine is ₹ 11,000 and per unit average cost of production is ₹ 5,500 (40% is general fixed overhead cost).

The Finance Officer has been asked to undertake a customer profitability analysis and calculate and compare the profit margin per customer (before deducting general fixed overhead) to know about the real customer profitability.

Following are the additional overhead information:

Delivery costs	₹ 200 per kilometer
Emergency delivery cost (in addition to delivery cost)	₹ 21,000 per delivery
Order processing cost	₹ 6,000 per order
Specific discount and sales commission	As per negotiation
Product Advertisement cost	Actual cost

The following data are available for each customer.

	MT Ltd.	KG Ltd.	MG Bros.
Sales (in units)	2,000	1,000	800
Total delivery kilometer travelled	1,000	800	900
No. of emergency delivery	2	1	0
No. of orders processed	4	2	8
Specific Discount (percentage of sales revenue)	25%	20%	15%
Sales commission (percentage of sales revenue)	15%	10%	5%
Advertisement costs (₹)	8,75,000	6,15,000	4,30,000

You are required to analyse the profitability for each customer, which customer is the most profitable.

Cost Accounting Systems

12. XYZ Limited manufactures a standard product in a competitive market. During the year, the company experienced operational changes, including partial automation and fluctuations in input costs. Although the company continued using traditional costing methods, discrepancies were observed between financial and cost profits. Management has requested a detailed analysis to understand the variance and assess the accuracy of overhead absorption.

The following is the Trading and Profit & Loss Account of XYZ Limited:

Dr.	₹	Cr.	₹	₹
To Raw Materials	32,50,000	By Sales (40,000 units)	92,00,000	
To Direct Wages	21,75,000	By Closing Finished Goods (2,000 units)	3,90,000	
To Production Overheads	11,20,000	By Work-in-Progress:		
To Administration Overheads	6,25,000	Materials	72,000	

To Selling & Distribution Overheads	4,60,000	Wages	34,000	
To Preliminary Expenses Written Off	30,000	Production Overheads	26,500	1,32,500
To Goodwill Written Off	55,000	By Dividend Received	5,25,000	
To Fines	10,000	By Interest from Bank Deposits	1,20,000	
To Interest on Term Loan	18,000			
To Loss on Sale of Equipment	20,000			
To Tax	2,25,000			
To Net Profit for the Year	23,79,500			
Total	1,03,67,500		Total	1,03,67,500

XYZ Limited manufactures a standard product and uses traditional costing systems for internal reporting.

The Cost Accounting records of XYZ Ltd. show the following:

1. Production Overheads have been absorbed into work-in-progress at 25% on Prime Cost.
2. Administration Overheads have been charged at ₹14.75 per unit of finished goods produced.
3. Selling and Distribution Overheads have been charged at ₹11.50 per unit sold.
4. The under- or over-absorption of overheads has not been adjusted in the Costing Profit & Loss Account.

You are required to:

1. Prepare a Proforma Costing Profit and Loss Account, showing the net profit as per cost accounts.

2. Prepare control accounts for:
   - Production Overheads
   - Administration Overheads
   - Selling & Distribution Overheads
3. Reconcile the profit as per cost accounts with the profit shown in the financial accounts.

Job & Batch Costing

13. A manufacturing company follows a job costing system. The following data is extracted from its books for the year ended 31st March, 2025:

Particulars	Amount (₹)
Direct Materials	12,50,000
Direct Wages	9,80,000
Selling & Distribution Overheads (Variable: 60%, Fixed: 40%)	6,30,000
Administration Overheads (Fixed)	5,40,000
Factory Overheads (70% Fixed, 30% Variable)	5,75,000
Profit	8,25,000

Additional Information:

1. **Overhead Recovery:**
   - Factory overheads are absorbed as a % of Direct Wages.
   - Selling & Distribution and Administration overheads are recovered as a % of Cost of Production.
2. In 2025-26, the company received a new job order. It is estimated that direct materials required will be ₹ 3,20,000 and direct labour will cost ₹ 2,10,000

Additional Information:

- Selling & Distribution overheads have increased by 20% (with variable portion now being 65%).
- Factory overheads have decreased by 10% due to efficiency improvements.

- The company wants to maintain the same profit % on sales as in the previous year.

You are required to prepare:

(a) Prepare a Job Cost Sheet showing Prime Cost, Cost of Production, Cost of Sales, Sales Value

(b) Determine the price to be quoted for the new job in 2025-26, considering Revised overhead structure and the company's profit objective.

(Assume relevant cost rates are based on previous year's data unless specified otherwise.)

Process Costing

14. Following data are available for a product for the month of April, 2025:

Particulars	Process- I (₹)	Process- II (₹)
Opening work-in- progress	Nil	Nil
Costs incurred during the month:		
- Direct materials	6,00,000	
- Labour	1,20,000	1,60,000
- Factory overheads	2,40,000	2,00,000
Units of production:		
Received in process	40,000	36,000
Completed and transferred	36,000	32,000
Closing work-in-progress	2,000	?
Normal loss in process	2,000	1,500

Production remaining in process has to be valued as follows:

Materials 100% Labour 50% Overheads 50%

There has been no abnormal loss in Process- II.

The company follows weighted average method for valuing inventory.

Prepare Process Accounts after working out the missing figures and with detailed workings.

Joint Product and By-Product

15. A company processes a raw material in its Department 1 to produce three products, viz. P, Q and C at the same split-off stage. During a period 1,80,000 kgs of raw materials were processed in Department 1 at a total cost of ₹ 12,88,000 and the resultant output of P, Q and C were 18,000 kgs, 10,000 kgs and 54,000 kgs respectively. P and Q were further processed in Department 2 at a cost of ₹1,80,000 and ₹1,50,000 respectively.

C was further processed in Department 3 at a cost of ₹1,08,000. There is no waste in further processing. The details of sales affected during the period were as under:

	P	Q	C
Quantity Sold (kgs.)	17,000	5,000	44,000
Sales Value (₹)	12,24,000	2,50,000	7,92,000

There were no opening stocks. If these products were sold at split-off stage, the selling prices of P, Q and C would have been ₹ 50, ₹ 40 and ₹ 10 per kg respectively.

Required:

(i) Prepare a statement showing the apportionment of joint costs to P, Q and C.

(ii) Present a statement showing the cost per kg of each product indicating joint cost and further processing cost and total cost separately.

(iii) Prepare a statement showing the product wise and total profit for the period.

(iv) State with supporting calculations as to whether any or all the products should be further processed or not

Standard Costing

16. A company produces a product X, using raw materials A and B. The standard mix of A and B is 1: 1 and the standard loss is 10% of input. You are required to COMPUTE the MISSING INFORMATION indicated by "?" based on the data given below:

	A	B	Total
Standard price of raw material (₹ / kg.)	24	30	
Actual input (kg.)	?	70	
Actual output (kg.)			?
Actual price ₹ / kg.	30	?	
Standard input quantity (kg.)	?	?	
Yield variance (sub usage)	?	?	270(A)
Mix variance	?	?	?
Usage variance	?	?	?
Price variance	?	?	?
Cost variance	0	?	1,300(A)

Marginal Costing

17. Jupiter Ltd, a 'Fast-Moving Consumer Goods (FMCG)' company intends to diversify the product line to achieve full utilisation of its plant capacity. As a result of considerable research made, the company has been able to develop a new product called 'EXE'.

'EXE' is packed in *cans* of 100 ml capacity and is sold to the wholesalers in cartons of 24 *cans* at ₹120 per carton. Since the company uses its spare capacity for the manufacture of 'EXE', no additional fixed expenses will be incurred. However, accountant has allocated a share of ₹1,12,500 per month as fixed expenses to be absorbed by 'EXE' as a fair share of the company's present fixed costs to the new product for costing purposes.

The company estimates the production and sale of 'EXE' at 1,50,000 *cans* per month and on this basis the following cost estimates (per carton) have been developed:

	₹
Direct Materials	54
Direct Wages	36
All Overheads	27
Total Costs	117

After a detailed market survey the economy is confident that the production and sales of 'EXE' can be increased to 1,75,000 *cans* per month and ultimately to 2,25,000 *cans* per month.

The company at present has a capacity for the manufacture of 1,50,000 empty *cans* and the cost of the empty *cans* if purchased from outside will result in a saving of 20% in direct material, 10% in direct wages and 10% in variable overhead costs of 'EXE'. The price at which the outside firm is willing to supply the empty *cans* is ₹0.675 per empty *can*. If the company desires to manufacture empty *cans* in excess of 1,50,000 *cans*, a machine involving an additional fixed overhead of ₹7,500 per month will have to be installed.

Required:

(a) State by showing your workings whether the company should make or buy the empty *cans* at each of the three volumes of production of 'EXE' namely, 1,50,000, 1,75,000 and 2,25,000 *cans*.

(b) At what volume of sales will it be economical for the company to install the additional equipment for the manufacture of empty cans?

(c) Evaluate the profitability on the sale of 'EXE' at each of the aforesaid three levels of output based on your decision and showing the cost of empty *cans* as a separate element of cost.

Budget and Budgetary Control

18. SIAM Ltd. manufactures two products using one type of material and one grade of labour. Shown below is an extract from the company's working papers for the next period's budget.

Particulars	Product A	Product B
Budgeted Sales (Units)	1,800	2,400

Budgeted Material Consumption per Product (Kg.)	5	3
[Budgeted Material Cost ₹12 per Kg.]		
Standard Hours Allowed per product	5	4
[Budgeted Wage Rate ₹8 per hour]		

Overtime premium is 50% and is payable, if a worker works for more than 40 hours a week. There are 45 direct workers.

The target productivity ratio (or efficiency ratio) for the productive hours worked by the direct workers in actually manufacturing the products is 80%; in addition the non-productive downtime is budgeted at 20% of the productive hours worked.

There are twelve 5-day weeks in the budget period and it is anticipated that sales and production will occur evenly throughout the whole period.

It is anticipated that stock at the beginning of the period will be:

Product A 510 units; Product B 1,200 units; Raw material 2,150 Kg.

The target closing stock, expressed in terms of anticipated activity during budget period are - Product *A* 15 days sales; Product *B* 20 days sales; Raw material 10 days consumption.

Required:

(a) the material purchases budget, and

(b) the wages budget for the direct workers, showing the quantities and values, for the next period.

Miscellaneous

19. (a) Standard costs, marginal costs are some of the types of cost which helps the management in decision making. DISCUSS any five types of costs categorised based on its use in Managerial Decision Making.

(b) Provide EXAMPLE(S) of the cost driver for the following cost pools:

Cost Pool
Quality Control

Research and Development
Machine Maintenance
Employee Training Costs
Customer Service

(c) Besides having advantages of Budgetary Control System being a powerful instrument used by business entity for the control of their expenditure, it has certain limitations as well. ELABORATE any four limitations of Budgetary Control System.

(d) Cost control emphasis is on past and present, while cost reduction emphasis is on present and future. DISCUSS some more differences between Cost control and Cost reduction.